

Assessment of Spatio-Temporal Vegetation Dynamics of Himachal Pradesh Using MODIS Time Series Vegetation Indices

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Declaration: Neither the whole article nor any substantial portion of it has been published elsewhere, it is the original work by the authors.

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Abstract:

This study examined the spatio-temporal dynamics of vegetation to identify and characterize vegetation stress zones in Himachal Pradesh, India, utilizing the Normalized Difference Vegetation Index (NDVI) and Vegetation Condition Index (VCI). The indices were obtained from time-series MODIS 16-day composite products at a spatial resolution of 250 m for all months from 2015 to 2022. Monthly NDVI layers were aggregated to obtain the yearly mean NDVI, and a long-term NDVI baseline was established by synthesizing multi-year NDVI composites. The Vegetation Condition Index (VCI) was calculated utilizing both long-term and current Normalized Difference Vegetation Index (NDVI) measurements and subsequently categorized into four classifications: High, Moderate, Low, and No Vegetation. The spatial expanse of each class was measured in square kilometers and expressed as a percentage. Data processing and analysis

utilized ArcGIS 10.4, ERDAS Imagine, Notepad++, and MS Excel. The variability of VCI was assessed year-wise by a comparison analysis utilizing bar graphs. The findings indicate significant interannual variability in vegetation condition, with 2015 showing the least area under moderate and high VCI, signifying extensive vegetation stress, while 2021 demonstrated the greatest extent of these categories, indicating favorable vegetation conditions due to enhanced climatic and monsoonal patterns. The research findings show that the effectiveness of multi-temporal MODIS-based VCI analysis in assessing vegetation health and identifying environmental stress in mountainous environment.

Keywords: Himachal Pradesh, MODIS, NDVI, VCI, Vegetation.

Introduction:

Ecology is described as the study of the interdependent relationships between organisms and their environment. It pertains to the hierarchical systems above organisms within the biological spectrum (Fundamentals of Ecology). The term "ecosystem" was initially introduced by Sir Arthur Tansley of Britain in 1935. An ecosystem is an abstract term relevant to any system that encompasses a creature. Ecological studies fundamentally examines organisms and their surrounding environment, encompassing both biotic and abiotic components. Biodiversity pertains to the diversity of biological forms and ecological systems (Remote Sensing of the Environment: An Earth Resource Perspective) (Huttich et al., 2011). Ecosystems are dynamically interacting systems comprising animals, the communities they form, and the abiotic components of their environment. Ecosystem activities, including primary production, nutrient cycling, and niche creation, govern the flow of energy and matter within an environment. Intensive and illogical human activities have significantly jeopardized natural ecosystems in recent decades (Bezerra et al., 2020). Throughout natural evolution, terrestrial ecosystems have endured the demands of

human activity, leading to the disruption of ecological balance. The degradation of the natural environment significantly impacts human life and sustainable development (Fang et al., 2014). Alterations in vegetation serve as effective markers for ecological and environmental assessments at both local and global stages (Sobrino and Julien 2011; Schucknecht et al. 2013; Guay et al. 2014). The growing seasons indicate that NDVI and VCI were correlated with the mean 1-month SPI during the growing season for dry (2003) and wet (2007) conditions, reflecting spatio-temporal variation in the analysis (Reddy et al., 2020).

To estimate above ground plant biomass, Lumbierres et al., (2017) used NDVI Land Surface Phenology (LSP) from MODIS, calibrated and validated in Doñana National Park's marsh vegetation. LSP-Maximum-NDVI correlated well with ground-truth biomass production data from 2001-2015 ($R^2 = 0.65$). Single-image MODIS NDVI estimators performed badly ($R^2 < 0.41$). The LSP-Maximum-NDVI estimator can tolerate precipitation, hydroperiod, and plant community composition changes. Remote-sensing methods that reliably quantify plant biomass and ground-truth data may help manage seasonal marsh habitats. (Qin et al., 2015) mapped Chinese forests using 2010 MODIS time series data and ALOS PALSAR orthorectified 50-m mosaic pictures (FBD mode with HH and HV polarisation). To conceal built-up, barren, and poorly vegetated areas, they developed an annual maximum NDVI map using MODIS-based NDVI dataset (MOD13Q1, 250-m spatial resolution). They built a decision tree method to distinguish forest from non-forest land cover using PALSAR backscatter coefficient data signature analysis. PALSAR mapped China's forests in 2010. Forest/non-forest classification map was 96.2% accurate and 0.91 Kappa. The 50-m PALSAR forest cover map was compared to five databases. PALSAR-based forest map estimations of China's total forest area (2.02 106 km²) match China National Forestry Inventory (1.95 km²), JAXA (2.00 km²), and FAO FRA (2.07 km²). The PALSAR-based forest map exhibits good linear connections with JAXA, MCD12Q1,

and NLCD-China datasets at province and county levels. All forest maps feature comparable pixel-scale forest/non-forest distributions. Our PALSAR forest map pinpoints Chinese agro-forests. This study indicates that PALSAR-MODIS images can map huge woods. Forest cover map results.

Vulnerability denotes both susceptibility to a threat and the possible degradation of environmental assets or value due to that threat (Hulley et al., 2019). Vulnerability refers to the degree to which a community, structure, service, or geographical area is susceptible to damage or disruption from specific dangers. Satellite data, together with GIS and remote sensing techniques, is a crucial component in the study of forest ecosystems (Verbesselt et al., 2009). Modis satellite imagery offers comprehensive and regular coverage of wide regions, enabling the monitoring of forest health. In recent years, the use of time series remote sensing datasets and geographic information systems (GIS) has escalated in the evaluation of spatio-temporal patterns of vegetation dynamics at regional to global scales (Zhou et al. 2001; Cleland et al. 2006; Xie et al. 2015; Cao et al. 2018). The pronounced disparity between red and near-infrared reflectance facilitates the creation of vegetation indices such as NDVI and VCI (Ha et al., 2024). This work utilises MODIS-based NDVI data (MOD 13 Q1.v 006, 250 m spatial resolution) for the creation of annual NDVI. The annual NDVI for each year is aggregated to create a long-term NDVI, followed by the preparation of a VCI map for each year. The model maker in Erdas Imagine 2014 was utilised to compute the stack mean from stacked photos. MODIS data is crucial in assessing forest and ecosystem vulnerability, featuring a high temporal resolution of 16 days and a spatial resolution of 250 meters. The primary objective of the study is to evaluate spatio-temporal vegetation dynamics utilising MODIS (MOD 13 Q1 v006, 250m) data, and to generate NDVI and Vegetation Condition Index (VCI) maps for each year, while calculating the annual area and percentage across various VCI classifications.

Study Area:

Himachal Pradesh shares its northern boundaries with the union territories of Jammu & Kashmir and Ladakh, and to the east, it adjoins China. The state is bordered to the west by Punjab, to the southwest by Haryana, to the southeast by Uttarakhand, and to the south by Uttar Pradesh, creating a narrow boundary (Figure 1). Himachal Pradesh is situated between 30°22'N and 33°12'N latitude and 75°47'E and 79°04'E longitude, including an area of 55,673 square kilometers. This strategic location situates the state at the intersection of vital natural zones, enhancing its environmental, economic, and geopolitical significance (Chandel, 2025). The province is traversed by the Sutlej, Beas, Ravi, Chenab, and Yamuna rivers. The Himalayan River system is not as abundantly exploited for irrigation as it is in the plains; however, it serves as a water source for the Indus River basin. The utility of these rivers for irrigation is limited by the undulating terrain. The rivers experience significant flows during the rainy season, while the water is frozen at high altitudes due to snowfall in the winter. They contract into narrow streams. Nevertheless, these rivers offer a significant amount of potential for the production of hydroelectric power (Pandey, 2010). The climate of Himachal Pradesh ranges from semi-tropical to semi-arctic based on height. It comprises three seasons that influence economic development. The wet season extends from July to September, winter from October to March, and summer from April to June. In summer, there is an inflow of tourists to the state from both domestic and international sources (Pandey et al., 2015). The state's rainfall is predominantly affected by the Southwest monsoon (July–October) and the western disturbances (November–April). The study area comprises distinct zones, namely zone-I (Dhaulakuan 30° 04' N, 75° 05' E, 700 m above mean sea level), zone-II (Palampur 32° 06' N, 76° 03' E, 700–1500 m above mean sea level), and zone-III (Bajaura 31° 08' N, 77° E, 800–1200 m above mean sea level). Zone-I (low hills, subtropical zone) experiences hot summers, mild winters, and moderate rainfall. In contrast, Zone-II (mid hills, sub-humid zone) receives moderate

to high annual rainfall, spanning from 1200 to 1800 mm, with a well-distributed monsoon season (Pokhariyal et al. 2021; Kumar et al., 2025).

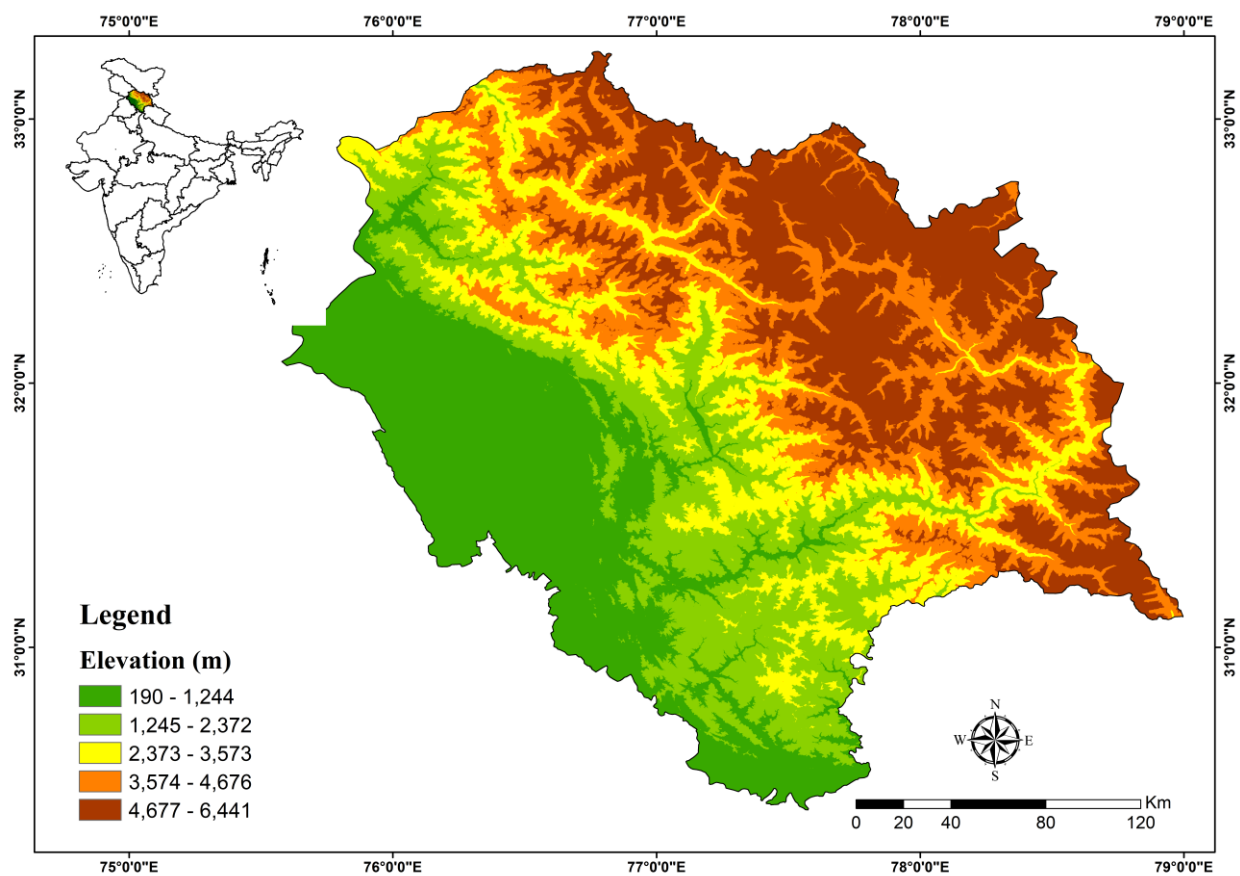


Figure 1. Location Map

Material:

This study employed NDVI and VCI data from the MODIS (MOD 13 Q1 v006, 250m) dataset, covering a decade (2015 to 2024), to generate monthly NDVI and VCI time series for vegetation

monitoring (Dabrowska-Zielinska et al. 2002; Singh et al. 2003). as shown in Table 1. This study utilizes a 10-year dataset of MODIS 16-day composite (MOD 13 Q1 v006) at a spatial resolution of 250 meters to prepare NDVI and VCI maps. It comprises 23 photos over the course of one year. The MODIS data was downloaded from USGS Earth Explorer. MODIS data was downloaded from NASA portal (<https://appeears.earthdatacloud.nasa.gov/>). Carto-sat DEM data is used for preparation of Slope and Elevation map of Study region which was downloaded from **USGS Earth explorer website (<https://earthexplorer.usgs.gov/>)**. **Softwares used** – For carrying out this work we used Arc GIS – 10.4, Q-GIS, ERDAS Imagine – 2015, Microsoft Excel, Jupiter Notebook, Notepad ++, R-Programming. This study employed NDVI and VCI data from the MODIS (MOD 13 Q1 v006, 250m) dataset, covering a decade (2015 to 2024), to generate monthly NDVI and VCI time series for vegetation monitoring (Dabrowska-Zielinska et al. 2002; Singh et al. 2003). MODIS MOD13Q1 (Version 006) NDVI time-series data, possessing a spatial resolution of 250 m, were utilized to assess spatio-temporal vegetation changes in Himachal Pradesh. The MODIS HDF files were converted to IMG format, reprojected from sinusoidal projection to the geographic coordinate system, and spatially clipped to the study area. Monthly NDVI composites were generated by aggregating bi-monthly NDVI images and computing their mean values. Annual NDVI layers were created by averaging monthly composites to establish long-term NDVI reference conditions. The Vegetation Condition Index (VCI) was calculated annually to evaluate vegetation health and inter-annual fluctuations. The area statistics of various VCI categories were analyzed to examine spatial patterns of vegetation condition across the region.

Table 1: Data Type and Sources

SNo.	Data Type	Source
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1.	MODIS (MOD 13 Q1 v006, 250m) data	NASA Portal (https://appears.earthdatacloud.nasa.gov/)
2.	Carto-sat DEM data	USGS Earth explorer website (https://earthexplorer.usgs.gov/).

Methods:

Figure 2 shows the procedure followed while carrying out this research work:

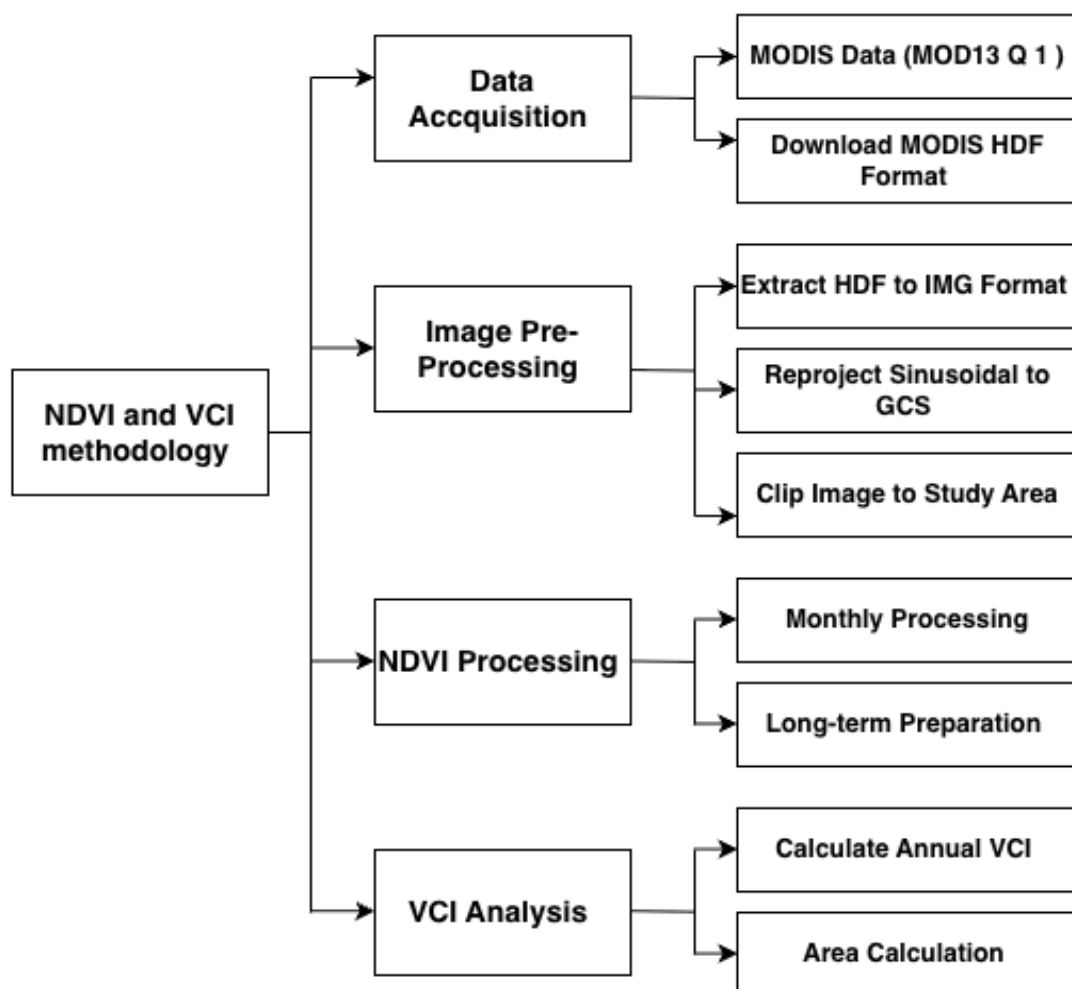


Figure 2. Flowchart of Methodology

Pre-Processing:

This study utilizes a 7-year dataset of MODIS 16-day composite (MOD 13 Q1 v006) at a spatial resolution of 250 meters to prepare NDVI and VCI maps. It comprises 23 photos over the course of one year. Monthly photos were aggregated, and subsequently, the stack mean was computed using ERDAS Imagine software. Throughout the month, photos were aggregated and analyzed to compute long-term NDVI. Then, the map is prepared using the Vegetation Condition Index (VCI) calculation. Carto-sat DEM data is utilized for the creation of slope and elevation maps of the study area. The study analyzed MODIS data of Himachal Pradesh, which was retrieved from NASA portal (<https://appeears.earthdatacloud.nasa.gov/>). The study's methodology is separated into three components: data pre-processing, data processing, and the analysis of processed data. NDVI, recognized as a premier indicator of vegetation coverage and greenness (Li et al. 2021), has been extensively utilized in evaluating ecological environments (Cai et al., 2015; Zeng et al., 2011), extracting vegetation phenology (Hmimina et al., 2013; Vrieling et al., 2018), and monitoring land cover dynamics (Baeza and Paruelo, 2020; Huang et al., 2017). The primary constraint of utilizing Vegetation Indices (VIs), especially those derived from the red and near-infrared segments of the electromagnetic spectrum, is their tendency to asymptotically achieve a saturation threshold at the peak of the season following the attainment of elevated biomass density (e.g., $NDVI > 0.7$, Huete et al., 2002). NDVI is calculated using the Equation (1):

$$NDVI = (NIR - Red) / (NIR + Red) \dots\dots\dots(1)$$

Where,

NIR = Near infrared reflectance, R= Red

This stage involves downloading data in HDF format with a spatial resolution of 250 m, followed by the conversion of the retrieved HDF data to IMG format in ERDAS Imagine, and subsequently reprojecting the data from Sinusoidal Projection to GCS Projection. Then the image is clipped as per the study area, A succinct overview of each dataset utilized in this investigation is presented in Table 1 and further detailed in that section.

Processing:

The Vegetation Condition Index (VCI) is derived from an adjustment to the minimal historical vegetation index value in relation to the Normalized Difference Vegetation Index (NDVI). The VCI compares the present Vegetation Index (VI), including NDVI or Enhanced Vegetation Index (EVI), with historical values from the same pixel over the corresponding period in previous years (Orimoloye et al., 2021). All reprojected files for each month across all years were stacked using Model Maker to generate the monthly NDVI mean. Similarly, the monthly NDVI data stacks were utilized to produce the Mean Annual NDVI data for each year, which was subsequently clipped using the Himachal Pradesh Shapefile. The mean NDVI and VCI profiles for the growing season were studied to evaluate their spatio-temporal vegetation dynamics from 2015 to 2022. A comprehensive examination of the growth season mean NDVI and VCI was conducted alongside the 1-month SPI for both dry (2016) and rainy (2019) years (Reddy et al., 2020). The Annual NDVI for all years was subsequently stacked in ERDAS Imagine (2014) for the further computation of VCI. VCI provides the deviation of each pixel from the historical NDVI values. The VCI assesses the health of vegetation, as well as its greenness, by taking into account a variety of stressors that are induced by environmental factors (Kogan, 1990; Liu and Kogan, 1996). VCI value is being measured in percentage ranging from 1 to 100. VCI value below 35% indicates severe drought condition, 35 to 50% shows the drought condition, and 50 to 100% indicates above normal condition of vegetation (Kogan 1995; Reddy et al., 2020). MODIS MOD13Q1 (Version 006) NDVI time-series data, with a spatial resolution of 250 m, were employed to evaluate spatio-

temporal vegetation dynamics in Himachal Pradesh. The MODIS HDF files were transformed into IMG format, reprojected from sinusoidal projection to the geographic coordinate system, and spatially clipped to the study area. Monthly NDVI composites were created by aggregating bi-monthly NDVI pictures and calculating their mean values. Annual NDVI layers were generated by averaging monthly composites to produce long-term NDVI reference conditions. The Vegetation Condition Index (VCI) was computed annually to assess vegetation health and inter-annual variations. The area statistics of several VCI categories were studied to investigate spatial patterns of vegetation condition throughout the region. The VCI is calculated per pixel. VCI is calculated using Equation (2):

$$VCI = [NDVI_{(current\ year)} - Stack_{MIN}] / [Stack_{MAX} - Stack_{MIN}] \dots\dots\dots(2)$$

Where,

$NDVI_{(current\ year)}$ – Particular year NDVI

$NDVI_{(max)}$ – Maximum NDVI values in all years NDVI

$NDVI_{(min)}$ – Minimum NDVI values in all years.

VCI is classified into four classes: High VCI, Moderate VCI, Low VCI and No Vegetation through Arc GIS – 10.4 Software.

Intra-year variations in VCI classes were evaluated using pie charts and tabulated summaries for each year, while inter-year comparisons of VCI area percentages were visualized through bar graphs generated in Excel. The VCI computation tables included parameters such as VCI value, VCI range, pixel count, area covered (sq. km), and percentage area for each category. The area under each VCI class was derived by multiplying the corresponding pixel count by the pixel area. After generating annual VCI maps for 2015 through 2022, year-to-year variations were quantified

and compared using statistical tables and graphical representations. Finally, a comprehensive multi-year comparison was conducted, resulting in an integrated graph that depicts the long-term spatio-temporal variability of VCI across Himachal Pradesh from 2015 to 2022.

Results and Discussion:

Vegetation condition maps for the years 2015 to 2024 have been prepared separately and are detailed below:

I) Year 2015:

In 2015, the Vegetation Condition Map illustrates the current state of vegetation. In the 2015 VCI map, the High VCI class encompasses an area of 8,799.89 sq. km (15.8%), the Moderate VCI class covers 11,544 sq. km (20.73%), the Low VCI class spans 13,443.2 sq. km (24.14%), and the No vegetation class occupies 21,905.85 sq. km (39.33%), as illustrated in Table 2. and Figure 3.

Table 2. Vegetation Condition Index (2015)

VCI	VCI Range	Area Covered (Sq. Km)	Area in Percentage (%)
No Vegetation	0 – 0.25	21905.8	39.33
Low VCI	0.25 – 0.5	13443.2	24.14
Moderate VCI	0.5 – 0.75	11544	20.73
High VCI	0.75 - 1	8799.89	15.8

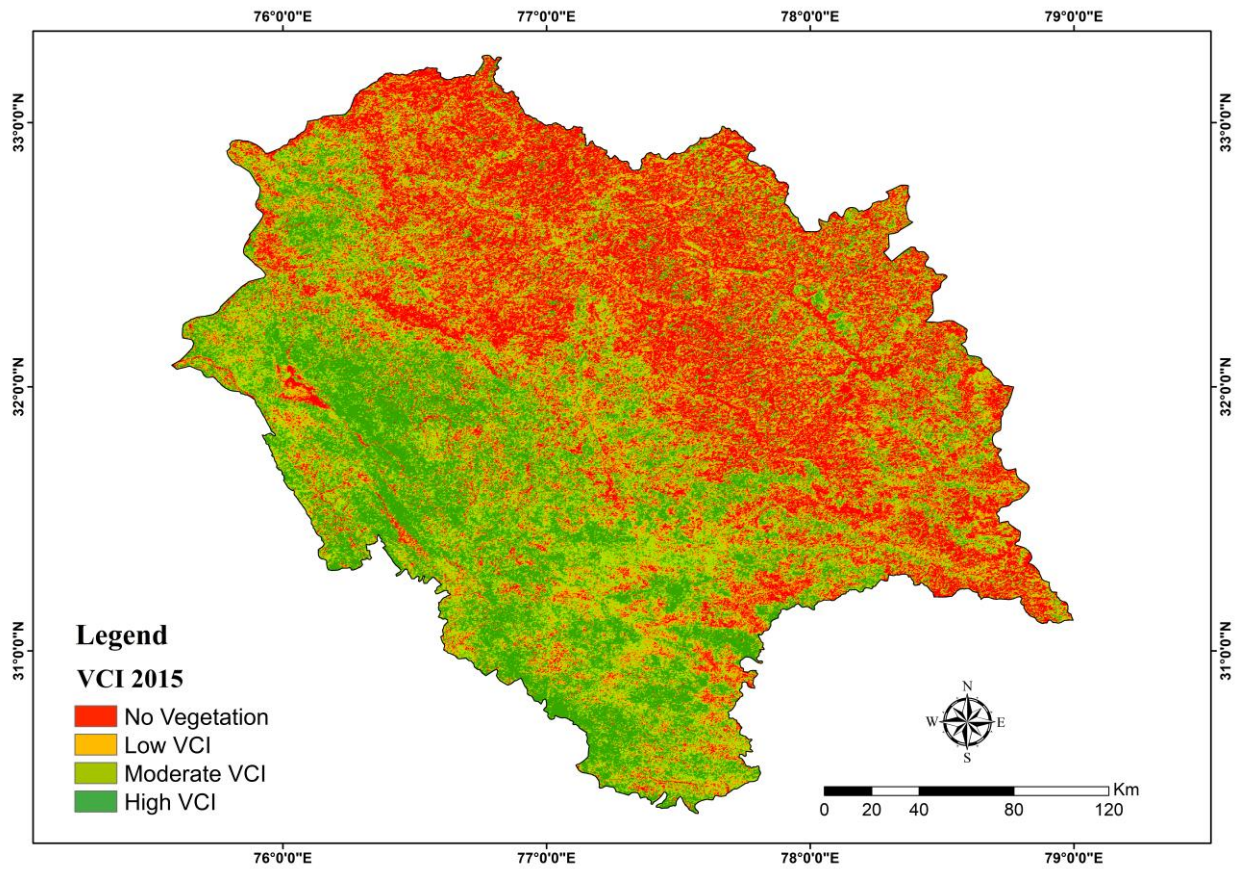


Figure. 3 Vegetation Condition Index (2015)

II) Year 2016:

In the 2016 VCI map, the High VCI class encompasses an area of 19,868.9 sq. km (35.68%), the Moderate VCI class covers 7,316.98 sq. km (13.14%), the Low VCI class spans 7,099.83 sq. km (12.75%), and the No Vegetation class occupies 21,407.2 sq. km (38.44%), as illustrated in Table 3 and Figure 4.

Table 3. Vegetation Condition Index (2016)

VCI	VCI Range	Area Covered (Sq. Km)	Area in Percentage (%)
No Vegetation	0 – 0.25	21407.2	38.44
Low VCI	0.25 – 0.5	7099.83	12.75

Moderate VCI	0.5 – 0.75	7316.98	13.14
High VCI	0.75 - 1	19868.9	35.68

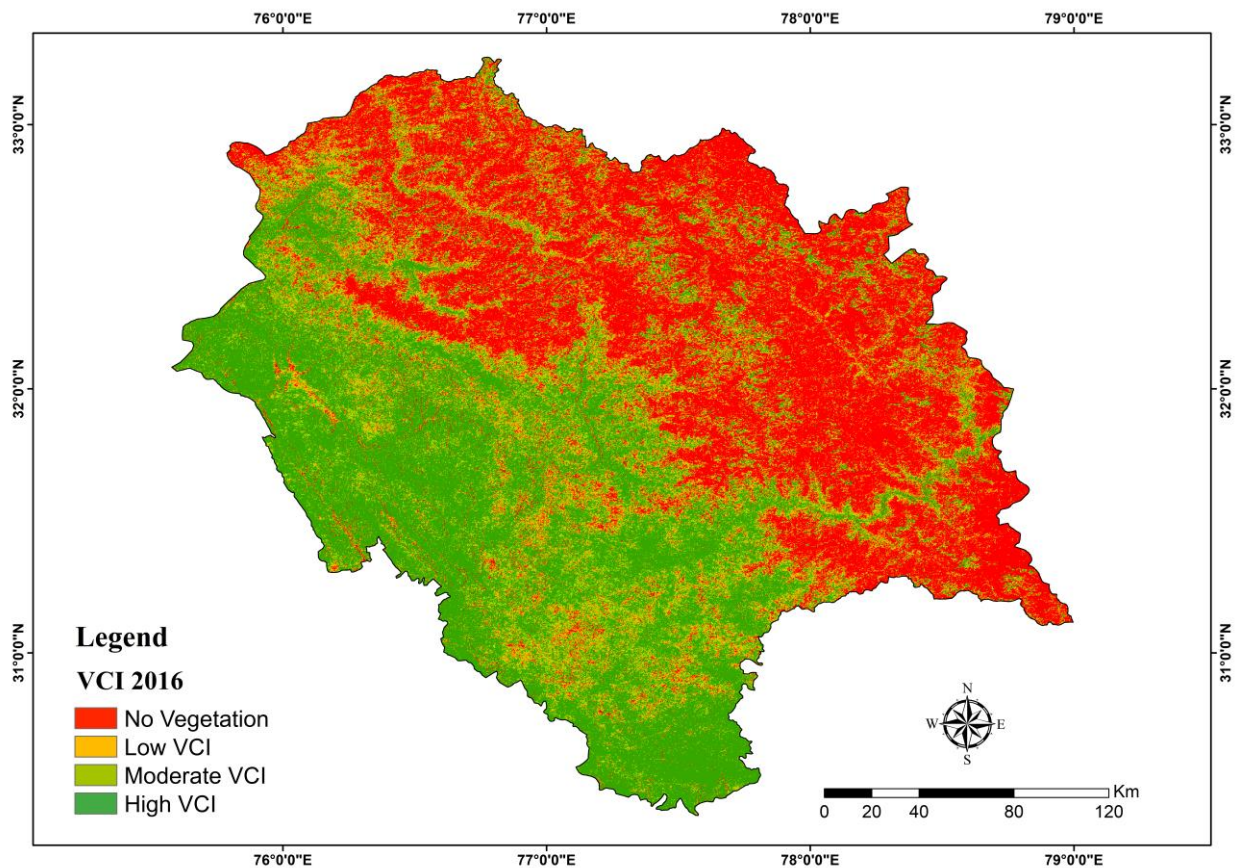


Figure 4. Vegetation Condition Map (2016)

III) Year 2017:

In the 2017 VCI map, the High VCI class encompasses an area of 20,161.9 sq. km (36.2%), the Moderate VCI class covers 14,720.2 sq. km (26.43%), the Low VCI class occupies 11,919.4 sq. km (21.4%), and the No Vegetation class accounts for 8,891.36 sq. km (15.97%), as illustrated in Table 4. and Figure 5.

Table 4. Vegetation Condition Index (2017)

VCI	VCI Range	Area Covered (Sq. Km)	Area in Percentage (%)
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No Vegetation	0 – 0.25	8891.36	15.97
Low VCI	0.25 – 0.5	11919.4	21.4
Moderate VCI	0.5 – 0.75	14720.2	26.43
High VCI	0.75 - 1	20161.9	36.2

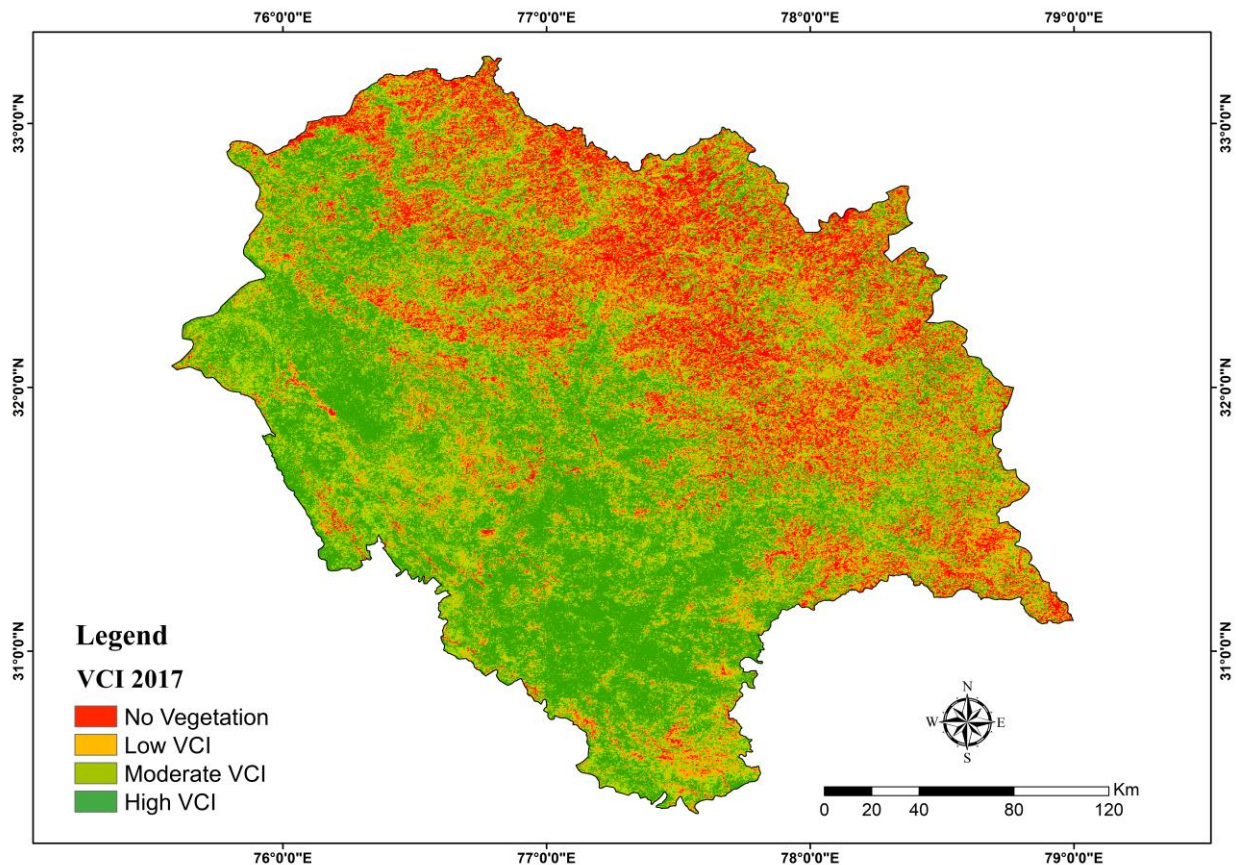


Figure 5. Vegetation Condition Map (2017)

IV) Year 2018:

In the 2018 VCI map, the High VCI class encompasses an area of 12,462.6 sq. km (22.38%), the Moderate VCI class covers 14,776.1 sq. km (26.53%), the Low VCI class spans 15,000.3 sq. km (26.93%), and the No Vegetation class occupies 13,453.9 sq. km (24.16%), as illustrated in Table 5. and Figure 6.

Table 5. Vegetation Condition Index (2018)

VCI	VCI Range	Area Covered (Sq. Km)	Area in Percentage (%)
No Vegetation	0 – 0.25	13453.9	24.16
Low VCI	0.25 – 0.5	15000.3	26.93
Moderate VCI	0.5 – 0.75	14776.1	26.53
High VCI	0.75 - 1	12462.6	22.38

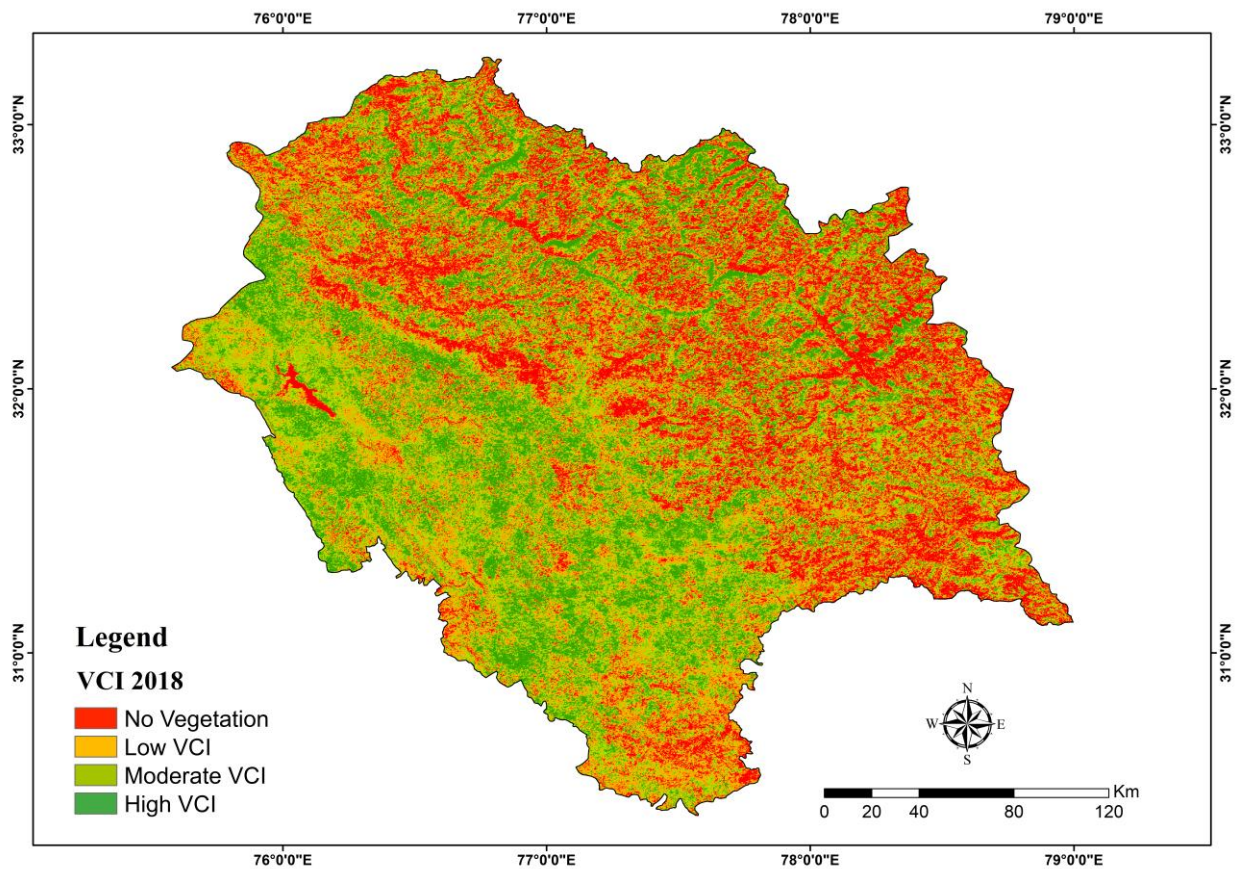


Figure 6. Vegetation Condition Map (2018)

V) Year 2019:

As illustrated in Table 6 and Figure 7, the 2019 VCI map exhibits a High VCI class covering 16136.5 sq. km (28.97%), a Moderate VCI class covering 9632.04 sq. km (17.29%), a Low VCI class covering 10257.4 sq. km (18.42%), and a No vegetation class covering 19667 sq. km (35.32%).

Table 6. Vegetation Condition Index (2019)

VCI	VCI Range	Area Covered (Sq. Km)	Area in Percentage (%)
No Vegetation	0 – 0.25	19667	35.32
Low VCI	0.25 – 0.5	10257.4	18.42
Moderate VCI	0.5 – 0.75	9632.04	17.29
High VCI	0.75 - 1	16136.5	28.97

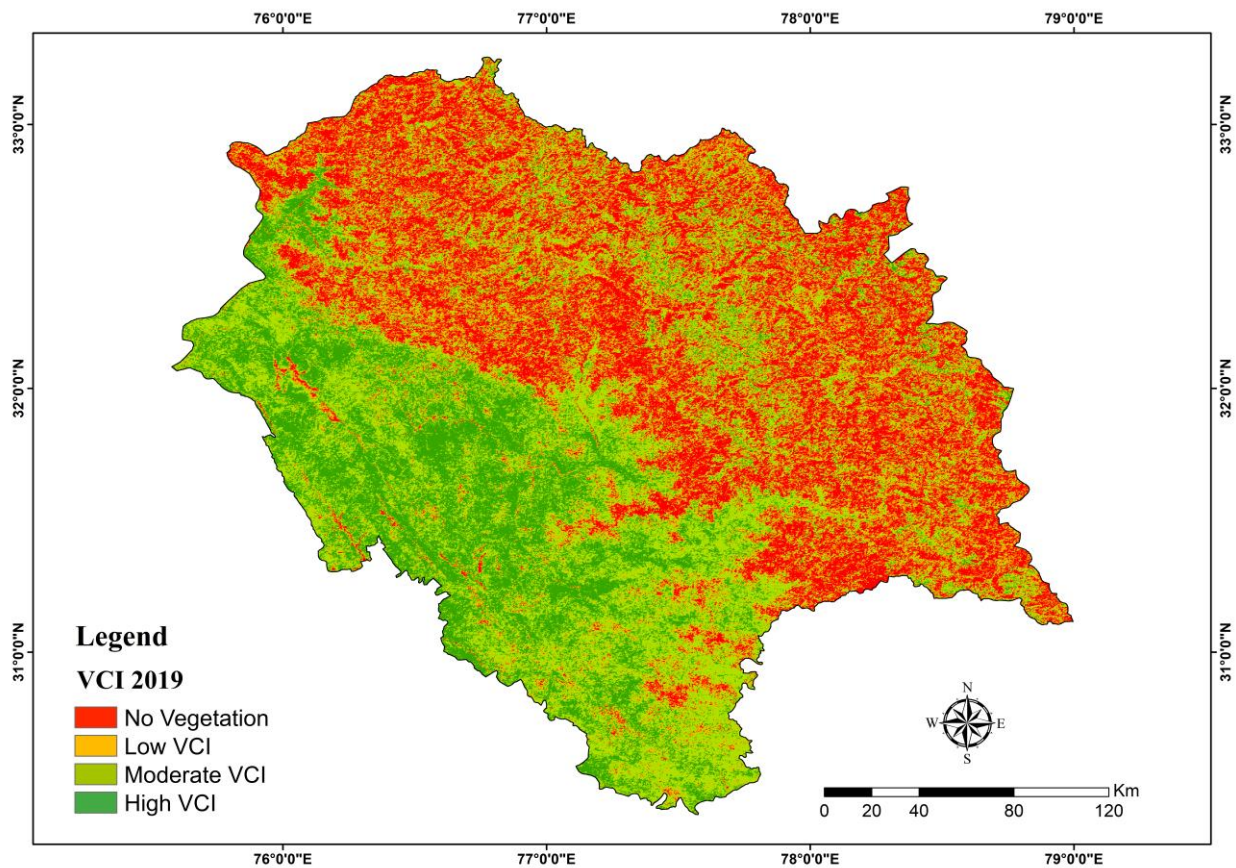


Figure 7. Vegetation Condition Map (2019)

VI) Year 2020:

In the 2020 VCI map, the High VCI class encompasses an area of 17,402.9 sq. km (31.225%), the Moderate VCI class covers 11,690 sq. km (20.99%), the Low VCI class spans

11,500.9 sq. km (20.65%), and the No Vegetation class occupies 15,099.1 sq. km (27.11%), as illustrated in Table 7 and Figure 8.

Table 7. Vegetation Condition Index (2020)

VCI	VCI Range	Area Covered (Sq. Km)	Area in Percentage (%)
No Vegetation	0 – 0.25	15099.1	27.11
Low VCI	0.25 – 0.5	11500.9	20.65
Moderate VCI	0.5 – 0.75	11690	20.99
High VCI	0.75 - 1	17402.9	31.25

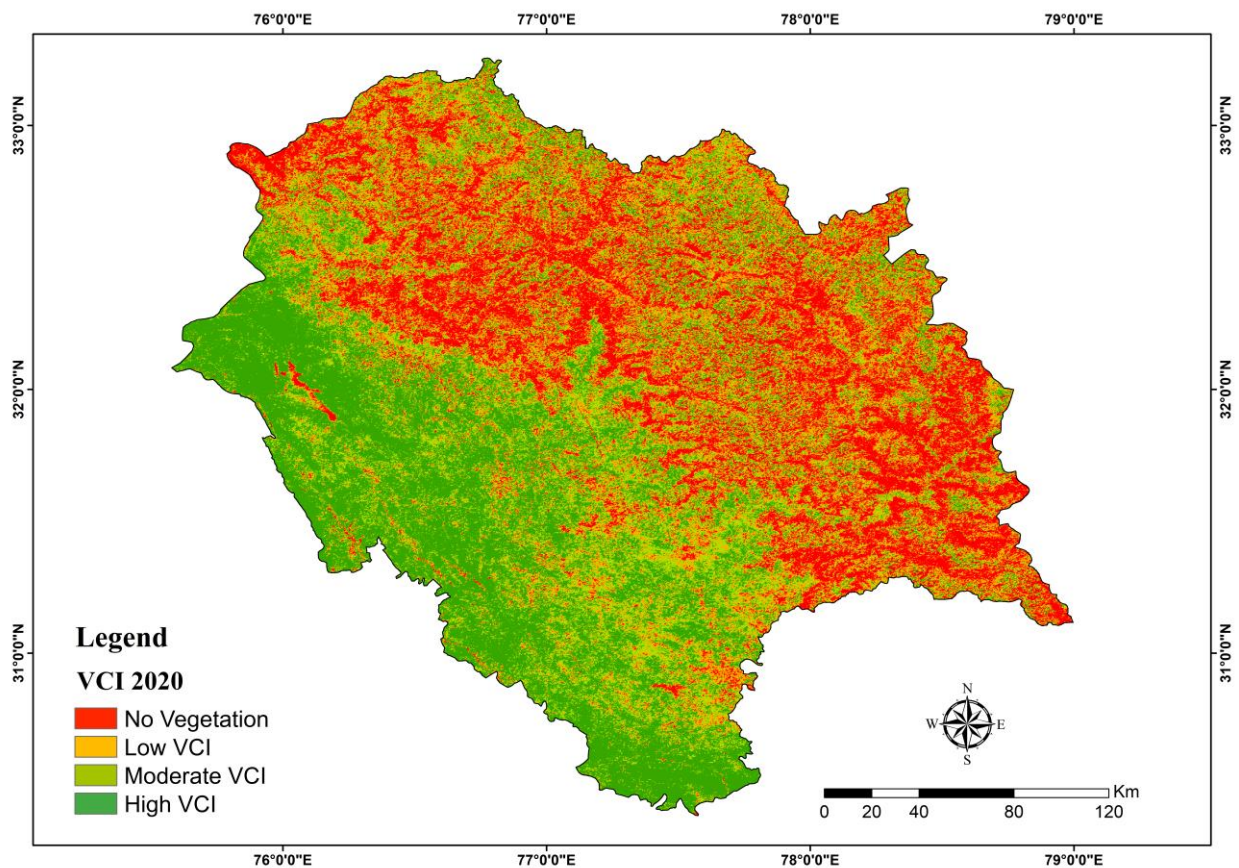


Figure 8. Vegetation Condition Map (2020)

VI) Year 2021:

In VCI map of 2021, High VCI class cover 12941.64 sq. km area (23.24 %), Moderate VCI Class cover 15532.77 sq.km area (27.89 %), Low VCI class cover 10160.22 sq.km area (18.24%) and No vegetation class cover 17058.23 sq.km area (30.63 %) as shown in table 8 and figure 9.

Table 8. Vegetation Condition Index (2021)

VCI	VCI Range	Area Covered (Sq. Km)	Area in Percentage (%)
No Vegetation	0 – 0.25	17058.23	30.63
Low VCI	0.25 – 0.5	10160.22	18.24
Moderate VCI	0.5 – 0.75	15532.77	27.89
High VCI	0.75 - 1	12941.64	23.24

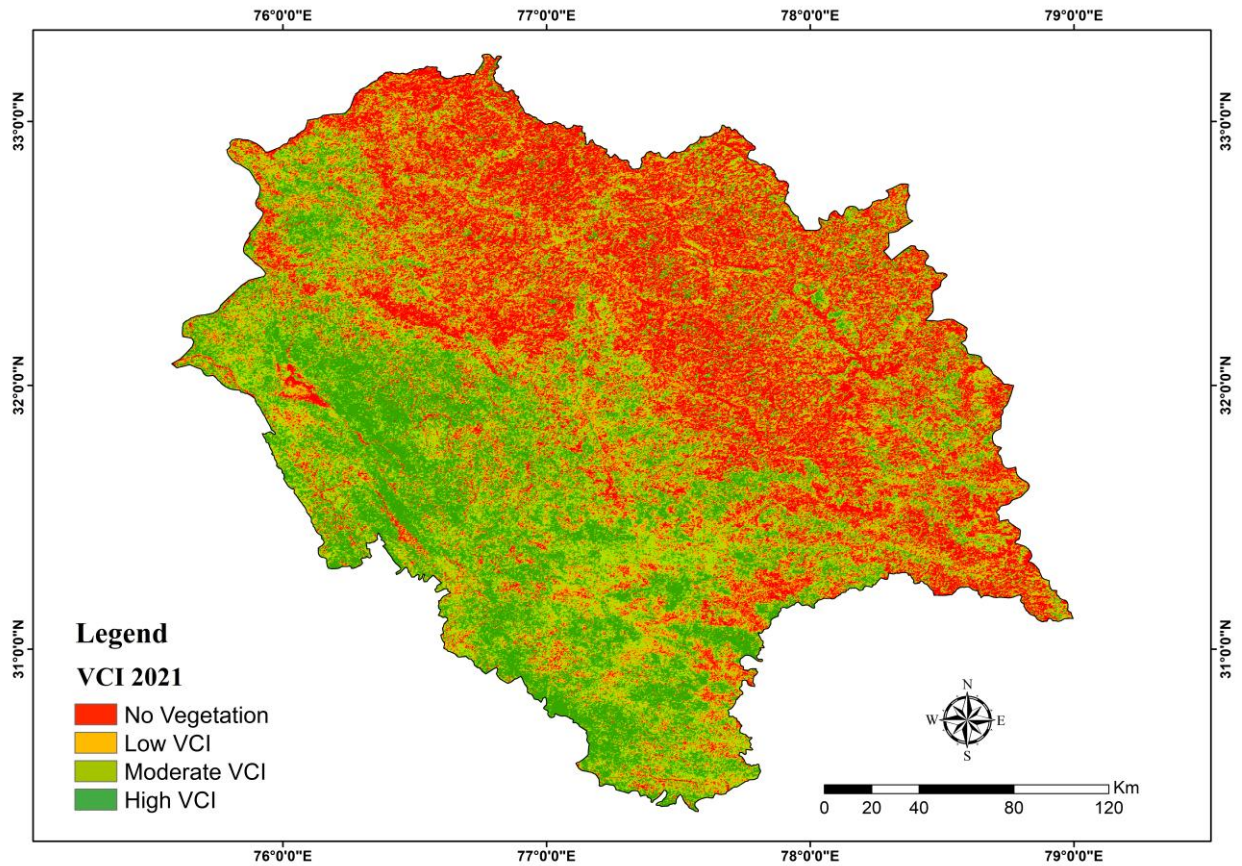


Figure 9. Vegetation Condition Map (2021)

VII) Year 2022:

In the 2022 VCI map, the High VCI class encompasses an area of 10,885.99 sq. km (19.55%), the Moderate VCI class covers 20,881.60 sq. km (37.49%), the Low VCI class spans 7,312.87 sq. km (13.13%), and the No Vegetation class occupies 16,612.39 sq. km (29.83%), as illustrated in Table 9 and Figure 10.

Table 9. Vegetation Condition Index (2022)

VCI	VCI Range	Area Covered (Sq. Km)	Area in Percentage (%)
No Vegetation	0 – 0.25	16612.39	29.83
Low VCI	0.25 – 0.5	7312.87	13.13
Moderate VCI	0.5 – 0.75	20881.60	37.49

High VCI	0.75 - 1	10885.99	19.55
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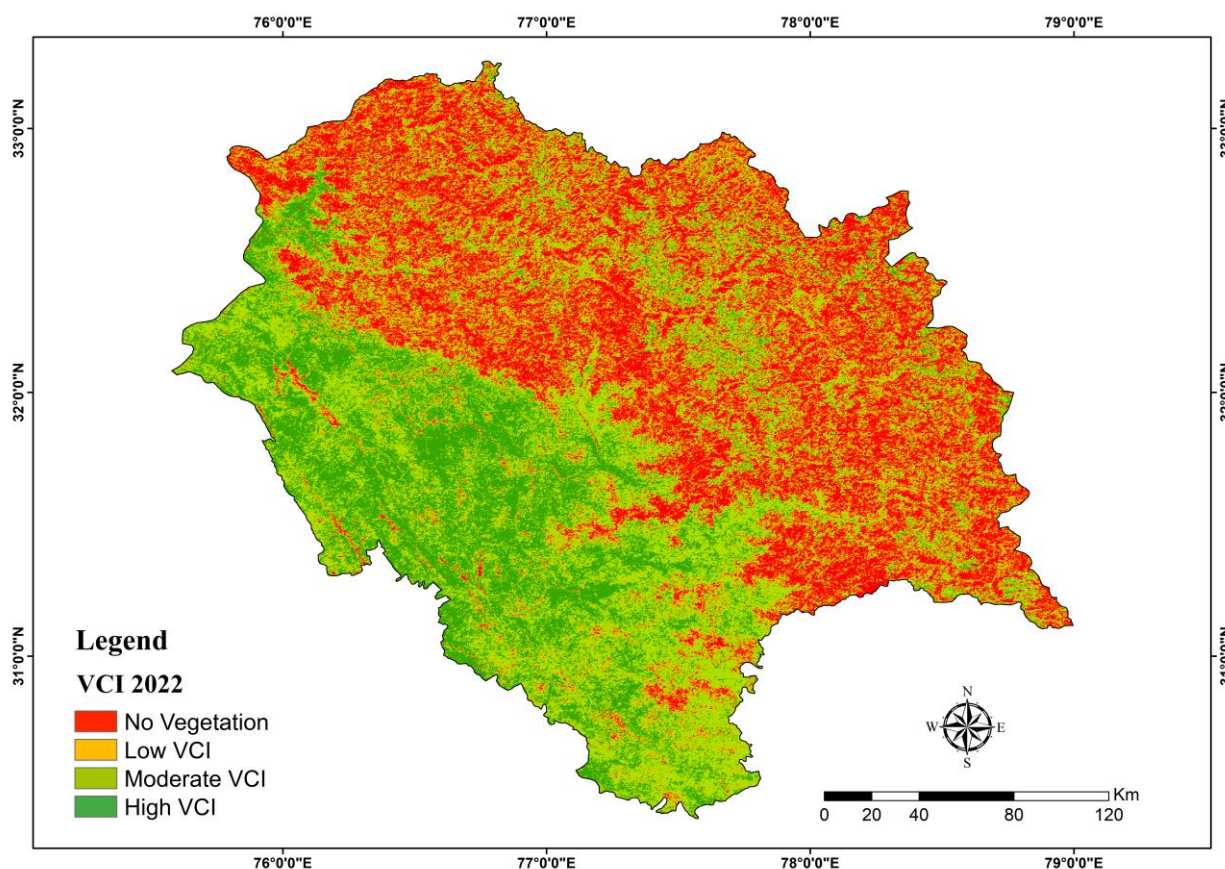


Figure 10. Vegetation Condition Map (2022)

Year wise VCI Comparison:

The graphic below illustrates the variations in No Vegetation, Low VCI, Moderate VCI, and High VCI from 2015 to 2022. The peak VCI occurred in 2021, followed by a decline in 2022. The No Vegetation index exhibits a slight diminishing trend over the eight-year period, while the Moderate VCI class shows a gradual and consistent increase. The Low VCI class likewise experienced a noticeable decline from 2015 to 2022, indicating a reduction in vegetation stress conditions. This suggests an overall augmentation of vegetation cover and improvement in vegetation health during the study period, reflecting a more favourable environmental impact. Figure 18 shows the overall

change in vegetation from 2015–2022. The increase in Moderate and High VCI classes further highlights improved plant vigor, likely influenced by better climatic and soil moisture conditions. These findings collectively indicate a positive ecological shift, driven by sustained vegetation growth over time. Figure 11. Shows the overall change in vegetation from 2015-2022.

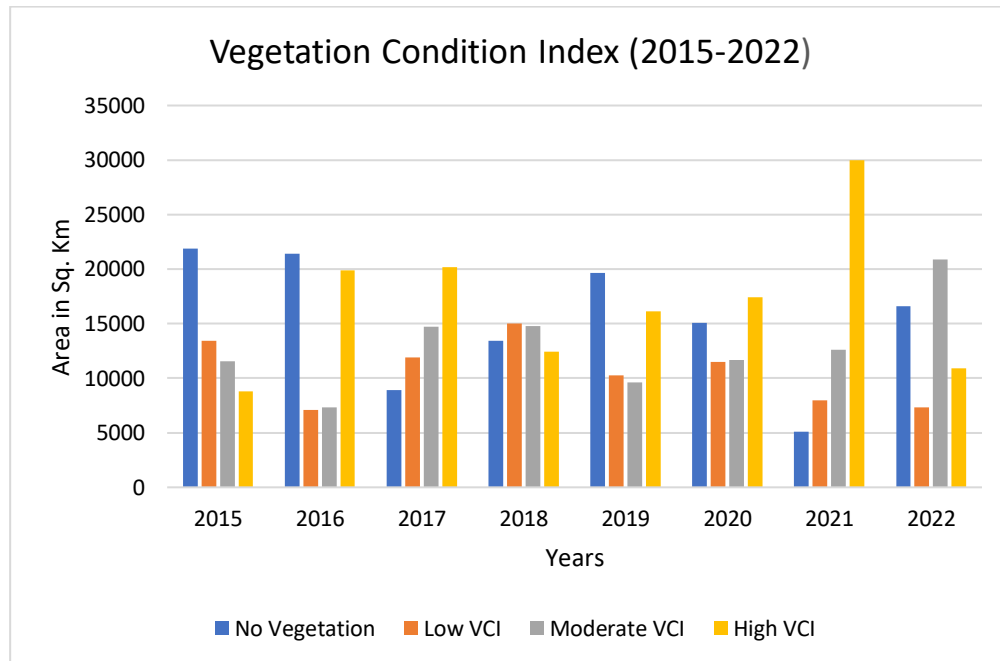
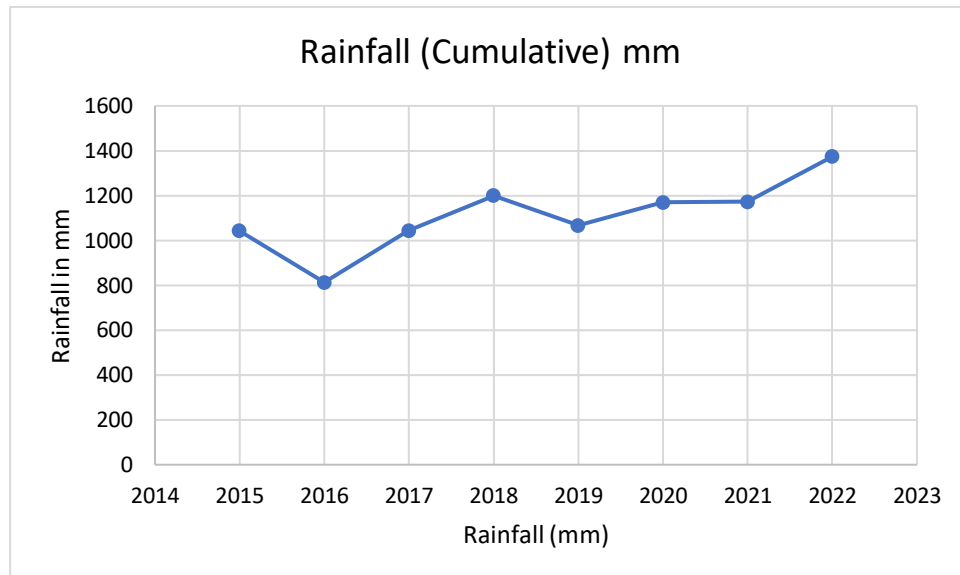


Figure 11. Vegetation Condition Map 2015-2022

Rainfall and VCI:

Figure 12 shows the cumulative rainfall in Himachal Pradesh from 2015 to 2022. The upward trend in Moderate and High VCI classes corresponds with the increase in yearly rainfall, especially post-2018, indicating that enhanced moisture availability facilitated healthier vegetation development. Years with increased cumulative rainfall, such as 2021 (1172.31 mm) and 2022 (1373.65 mm), correlate with heightened VCI values, signifying a positive association between rainfall distribution and vegetation vitality. Figure 19 Years with diminished rainfall, particularly 2016 (813.35 mm), indicate decreased vegetation performance, evidenced by increased proportions of Low VCI and No Vegetation categories, so affirming rainfall as a principal determinant of vegetation status. Table 17 The progressive reduction in Low VCI and No

vegetative regions from 2015 to 2022 can be ascribed to enhanced hydrological conditions, indicating that sustained and above-average precipitation boosts biomass productivity and overall vegetative resilience.



Source: JAXA/GPM_L3/GSMaP/v8/operational (GSMaP v8 - JAXA)

Figure 12. Yearly Cumulative Rainfall 2015-202

Conclusion

The VCI assessment for Himachal Pradesh from 2015 to 2022 indicates significant interannual fluctuations in vegetative condition, heavily influenced by monsoonal rainfall patterns throughout the state. Years with favorable precipitation, notably 2017 and 2021, demonstrate a significant increase in high-VCI regions, signifying augmented vegetation vitality and improved forest canopy conditions. In contrast, the years 2015, 2016, 2019, and 2022 demonstrate elevated levels of no-vegetation and low-VCI categories, indicative of the impact of subpar or unevenly distributed rainfall, which hindered vegetation recovery and diminished overall greenness. Years characterized by predominantly mild VCI dominance, such as 2018 and 2020, align with near-normal monsoon conditions that facilitated stable yet unremarkable vegetation growth. The multi-

year VCI trend indicates that vegetation and forest health in Himachal Pradesh are significantly affected by rainfall variability, with positive rainfall anomalies resulting in elevated VCI classes, whereas deficits or irregular monsoon periods cause diminished vegetation response.

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